Tutorial – Thin Airfoil Theory

Consider a NACA 23012 airfoil. The mean camber line for this airfoil is given by

$\bar{z} = 2.6595\left[\bar{x}^3 - 0.6075\bar{x}^2 + 0.1147\bar{x}\right]$ for $0 \le \bar{x} \le 0.2025$ and

$\bar{z} = 0.02208\left(1-\bar{x}\right)$ for $0.2025 \le \bar{x} \le 1.0$

Calculate the angle of attack at zero lift and the lift coefficient when the angle of attack is 4°.

Solution:

$\bar{x} = x/c$ and $\bar{z} = z/c$

To find the zero-lift angle of attack,

$$\alpha_{L=0} = -\frac{1}{\pi}\int_0^{\pi}\frac{dz}{dx}\left(\cos\theta - 1\right)d\theta$$

From the given mean camber line equation, we get

$$\frac{dz}{dx} = 2.6595\left[3\left(x/c\right)^2 - 1.215\left(x/c\right) + 0.1147\right] \quad \text{for } 0 \le \frac{x}{c} \le 0.2025$$

$$\frac{dz}{dx} = -0.02208 \quad \text{for } 0.2025 \le \frac{x}{c} \le 1.0$$

Transforming from cartesian coordinate (x) to polar coordinate (θ), we have

$$\frac{x}{c} = \frac{1}{2}\left(1-\cos\theta\right)$$

Therefore, we have

$$\frac{x}{c} = \frac{1}{2}\left(1-\cos\theta\right) \Rightarrow 0.2025 = \frac{1}{2}\left(1-\cos\theta\right) \Rightarrow 0.405 = 1-\cos\theta$$

$$\Rightarrow 0.595 = \cos\theta \Rightarrow \theta = 0.9335 \text{ rad}$$

For $0 \le \theta \le 0.9335$ rad

$$\frac{dz}{dx} = 2.6595\left[\frac{3}{4}\left(1-2\cos\theta+\cos^2\theta\right) - 0.6075\left(1-\cos\theta\right) + 0.1147\right]$$

$$\frac{dz}{dx} = 0.6840 - 2.3736\cos\theta + 1.995\cos^2\theta$$

For $0.9335 \text{ rad} \le \theta \le \pi$

$$\frac{dz}{dx} = -0.02208$$

For $0 \le \theta \le 0.9335$ rad

$$\frac{dz}{dx} = 0.6840 - 2.3736\cos\theta + 1.995\cos^2\theta$$

$$\frac{dz}{dx}(\cos\theta - 1) = -0.6840 + 3.0576\cos\theta - 4.3686\cos^2\theta + 1.995\cos^3\theta$$

For $0.9335 \text{ rad} \le \theta \le \pi$

$$\frac{dz}{dx} = -0.02208$$

$$\frac{dz}{dx}(\cos\theta - 1) = 0.02208 - 0.02208\cos\theta$$

Zero-lift angle of attack,

$$\alpha_{L=0} = -\frac{1}{\pi}\int_0^{\pi}\frac{dz}{dx}(\cos\theta - 1)\,d\theta$$

$$\int_0^{\pi}\frac{dz}{dx}(\cos\theta - 1)\,d\theta = \int_0^{0.9335}\frac{dz}{dx}(\cos\theta - 1)\,d\theta + \int_{0.9335}^{\pi}\frac{dz}{dx}(\cos\theta - 1)\,d\theta$$

From the table of integrals, we get

$$\int \cos\theta\,d\theta = \sin\theta$$

$$\int \cos^2\theta\,d\theta = \frac{1}{2}\sin\theta\cos\theta + \frac{1}{2}\theta$$

$$\int \cos^3\theta\,d\theta = \frac{1}{3}\sin\theta\left(\cos^2\theta + 2\right)$$

Substituting the above integrals, we get

$$\int_0^{0.9335}\frac{dz}{dx}(\cos\theta - 1)\,d\theta = \int_0^{0.9335}\left(-0.6840 + 3.0576\cos\theta - 4.3686\cos^2\theta + 1.995\cos^3\theta\right)d\theta$$

$$-0.6840\int d\theta = -0.6840\theta$$

$$+3.0576\int \cos\theta\,d\theta = 3.0576\sin\theta$$

$$-4.3686\int \cos^2\theta\,d\theta = -4.3686\left(\frac{1}{2}\sin\theta\cos\theta + \frac{1}{2}\theta\right) = -2.1843\sin\theta\cos\theta - 2.1843\theta$$

$$1.995\int \cos^3 d\theta = 1.995\left(\frac{1}{3}\sin\theta\cos^2\theta + \frac{2}{3}\sin\theta\right) = 0.665\sin\theta\cos^2\theta + 1.33\sin\theta$$

$$= \left[-2.8683\theta + 3.0576\sin\theta - 2.1843\sin\theta\cos\theta + 0.665\sin\theta\cos^2\theta + 1.33\sin\theta\right]_0^{0.9335}$$

$$\int_0^{0.9335} \frac{dz}{dx}(\cos\theta - 1)\,d\theta = -0.0065$$

$$\int_{0.9335}^{\pi} \frac{dz}{dx}(\cos\theta - 1)\,d\theta = \int_{0.9335}^{\pi}(0.02208 - 0.02208\cos\theta)d\theta$$

$$0.02208\int d\theta = 0.02208\theta$$

$$0.02208\int \cos\theta d\theta = 0.02208\sin\theta$$

$$\int_{0.9335}^{\pi} \frac{dz}{dx}(\cos\theta - 1)\,d\theta = \left[0.02208\theta - 0.02208\sin\theta\right]_{0.9335}^{\pi}$$

$$\int_{0.9335}^{\pi} \frac{dz}{dx}(\cos\theta - 1)\,d\theta = 0.0665$$

Zero-lift angle of attack,

$$\alpha_{L=0} = -\frac{1}{\pi}\left[-0.0065 + 0.0665\right] = -0.0191\text{ rad} = -1.09^\circ$$

The lift coefficient when the angle of attack is 4°,

$$\alpha = 4^\circ = 0.0698\text{ rad}$$

$$C_L = 2\pi(\alpha - \alpha_{L=0}) = 2\pi(0.0698 + 0.0191) = 0.559$$